

KAVINDA KEHELELLA

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RESEARCH INTERESTS

Natural Language Processing, LLM Robustness & Consistency, Knowledge Distillation in LLMs, Domain-Adapted Language Models, Explainability in AI

EDUCATION

University of Maryland, College Park, MD

Started Aug. 2023

Ph.D. in Electrical and Computer Engineering (CGPA: 3.92/4.00)

Advised by Prof. Sanghamitra Dutta

University of Moratuwa, Sri Lanka

Graduated May 2022

B.Sc. (Hons.), Electronic and Telecommunication Engineering, First Class (CGPA: 3.99/4.2)

Advised by Dr. Chamira Edussooriya, Dr. Ranga Rodrigo

Thesis: Radio frequency wireless sensing using micro-doppler signatures

PUBLICATIONS

- **K. Kehelella**, S. Dutta, “[COFFEQA: Counterfactual-Guided Few-Shot Distillation for LLM Question Answering](#),” Under Review at NeurIPS 2026.
- **K. Kehelella**, X. Wu, H. Ali, C. Yang, J. Xie, Y. Lee, R. Chen, S. Bhattacharyya “[RNDC-Lab: A Real-Time Calcium Imaging-Based Adaptive Neural Decoding Framework](#),” Accepted for publication in Frontiers in Neuroscience, 2026.
- **K. Kehelella**, G. Leelarathne, D. Marasinghe, N. Kariyawasam, V. Ariyaratna, A. Madanayake, R. Rodrigo, C. US Edussooriya, “[Vision Transformer with Convolutional Encoder-Decoder for Hand Gesture Recognition using 24 GHz Doppler Radar](#),” IEEE Sensors Letters, 2022.

RESEARCH EXPERIENCE

Counterfactual-Guided Few-Shot Distillation for LLM Question Answering 2026

Developed a few-shot knowledge distillation framework that incorporates factual examples with counterfactuals to improve knowledge transfer in LLMs for open-domain question answering. Introduced the Context Following Distillation (CFD) metric to validate generated counterfactuals based on teacher context dependence and answer agreement. Evaluated COFFEQA across eight QA benchmarks using T5 and Qwen2.5 teacher-student architectures, including analyses of teacher-student agreement and distillation near the teacher’s decision boundary.

Improving Consistency in LLM Question Answering

Ongoing

Investigated why LLMs produce inconsistent outputs on semantically equivalent, paraphrased questions by analyzing hidden-state representations layer-by-layer using Gemma-2B-it on HotpotQA. Trained probing classifiers to locate the attention heads most responsible for paraphrase-induced inconsistency, then applied targeted activation-editing interventions at those heads to reduce inconsistency. Built an evaluation pipeline reporting per-head and per-layer inconsistency deltas before and after editing, with aggregate improvement rates across evaluation instances.

A Unified Multimodal GNN-LLM Framework for Material Property Prediction Ongoing

Designed an embedding pipeline for biopolymer property prediction using domain-specific pretrained language models, ChemBERTa-2 and MatSciBERT. Engineered a substructure-to-text generation step that encodes per-component chemistry and inter-component interactions into natural-language descriptions. Combined structural and semantic embeddings by integrating domain-specific LLMs to improve downstream classification accuracy over a GNN-only baseline. Currently researching the implementation of an AI agent system to improve reasoning capabilities.

RNDC-Lab: Real-Time Calcium Imaging-Based Adaptive Neural Decoding

2025

Developing a real-time, end-to-end neural decoding system by integrating two main modules, NDSEP

and NeuroGRS, to improve suitability for real-time neural decoding. Tested the system on both simulated and real-world two-photon calcium imaging datasets.

Dynamically Reconfigurable Perception via Dataflow Parameterization 2024

Developed a parameterized method for pruning a large deep learning model for deployment in resource-constrained environments. The model is pruned during inference. Conducted experiments using ResNet50 with a Convolutional Block Attention Module on the ImageNet dataset.

Sentiment Transition Analysis in Updated Amazon Reviews 2024

Extracted and analyzed updated Amazon reviews across five product categories, comparing sentiment analysis approaches (GPT-4, a BERT-based sentiment classifier, VADER, and a fine-tuned DistilBERT) and highlighting the strengths and limitations of each.

Radio Frequency Wireless Sensing using Micro-Doppler Signatures 2022

Developed a Doppler radar system with directional sensing capability operating in the 2.45 GHz Industrial, Scientific, and Medical (ISM) band to recognize hand gestures in real time using machine learning, achieving more than 94% accuracy. The system includes the design and implementation of a patch antenna as the transmitter, an array of four patch antennas as the receiver, an analog multi-beam beamforming network with a Butler matrix, and backend software to recognize seven hand gestures. The proposed architecture consists of three modules: a convolutional encoder-decoder, an attention module with three transformer layers, and a multi-layer perceptron for real-time gesture recognition.

WORK EXPERIENCE

Software Engineer

May 2022 – Aug. 2023

MyBudget Holdings (Pvt) Ltd., Colombo, Sri Lanka

Designed a customer web portal using Go and Java; integrated a PostgreSQL/Kafka backend; maintained documentation via Confluence.

Intern Software Engineer

Oct. 2020 – Apr. 2021

Axiata Digital Labs, Colombo, Sri Lanka

Backend development with Springboot; implemented unit tests and worked within an agile team.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug. 2023 – Present

Dept. of Electrical and Computer Engineering, University of Maryland

ENEE440 Microprocessors; ENEE245 Digital Circuits and Systems Laboratory; ENEE408C/M Capstone Design Projects

Visiting Teaching Assistant

Jul. 2022 – Jul. 2023

Dept. of Electronic & Telecom. Engineering, University of Moratuwa

EN1094 Laboratory Practice; EN4720 Cyber Physical Systems

Skills

Languages: Python, C++, MATLAB, Java, Go, C

ML/NLP Frameworks: PyTorch, TensorFlow, HuggingFace, scikit-learn, Keras

Other Tools: Git, Jupyter, SQL (PostgreSQL, MySQL), JIRA, Confluence

HONORS AND AWARDS

- Graduate School Dean's Fellowship, University of Maryland (2023)
- Finalist, Hackstat 2.0 – ML/Data Science Competition, University of Colombo (2019)
- Finalist, SLIIT Datathon – Data Science Competition (2019)
- Finalist, Datastorm 1.0 – ML/Data Science Competition, University of Moratuwa (2019)
- Ranked 243rd (World), IEEE Xtreme 14.0 (2020)
- Ranked 489th (World), IEEE Xtreme 13.0 (2019)
- Mahapola Merit Scholarship, Government of Sri Lanka (2017)